MINI PROJECT

Distribution Analysis

Create a figure with TWO subplots:

- **Subplot 1:** Histogram with KDE for rental_count distribution
- **Subplot 2:** Boxplot for rental_count to show outliers

Categorical Distributions

Create a figure with TWO subplots:

- **Subplot 1: Count plot for season with proper ordering (Spring→Summer→Fall→Winter)**
- **Subplot 2: Count plot for events (exclude "none")**

Temperature Distribution

Create a violin plot showing the distribution of temperature_C for each season.

Temperature vs Rentals

Requirements:

- **Color points by season**
- **Add a regression line (linear fit)**
- **Size points by humidity_pct**
- **Add a legend with season labels**
- **Interpret: Is there a relationship? What's the optimal temperature for rentals?**

Categorical Comparisons

- **Subplot 1: Boxplot of rental_count by weather condition**
- **Subplot 2: Bar plot showing mean rental_count by events type (include confidence intervals/error bars)**

Time Series Analysis

Create a line plot showing rental_count over time (by date).

Requirements:

- **Add a 7-day rolling average (smooth line with different color)**
- **Highlight weekends (shade background or use different color markers)**

- **Add vertical lines or annotations for major events (optional)**
- **Interpret: Identify any trends or patterns in the time series**

Correlation Heatmap

Create a correlation heatmap for all numeric variables.

Requirements:

- **Use a diverging colormap (e.g., 'coolwarm')**
- **Display correlation values in the cells**
- **Add a title**
- **Interpret: Which variables are most strongly correlated with rental_count?**

Create a facet grid (pairplot or FacetGrid) showing the relationship between:

- **temperature_C vs rental_count**
- **Faceted by season and colored by weather**

Requirements:

- **Use Seaborn's FacetGrid or relplot**
- **Each facet should have a regression line**
- **Add appropriate labels**
- **Interpret: Compare the patterns across seasons**